

VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY



Topic: Industrial Production of 100 tonnes of Butyl Acetate

Subject- Fluid Mechanics

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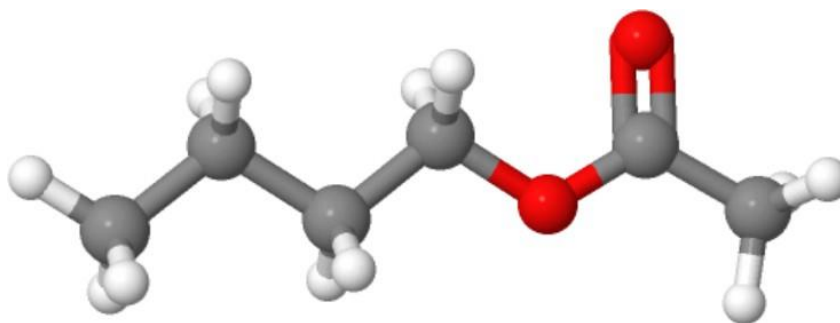
We would also like to thank our **seniors at VNIT, Nagpur**, for their helpful suggestions, constant motivation, and for sharing their experiences, which provided us with a clearer perspective and aided us in approaching the project more effectively.

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ABOUT BUTYL ACETATE



Chemical formula: $\text{CH}_3\text{CO}_2(\text{CH}_2)_3\text{CH}_3$

Reaction type: Esterification (Fischer–Speier esterification)

Reactants:

- Acetic acid (CH_3COOH)
- n-Butanol ($\text{CH}_3(\text{CH}_2)_3\text{OH}$)

Catalyst: Concentrated sulfuric acid (H_2SO_4) or p-toluenesulfonic acid (PTSA)

Reaction equation:



Conditions:

- Reflux at $\sim 120^\circ\text{C}$
- Removal of water drives the equilibrium to the product side (Le Chatelier's principle).

Occurrence in nature: Apples, especially of the Red Delicious variety, are flavored in part by this chemical. The alarm pheromones emitted by the Koschevnikov gland of honey bees contain butyl acetate.

Uses:

1. Paints, Coatings, and Lacquers: Butyl acetate is a primary solvent in the paint and coatings industry because it dissolves resins such as nitrocellulose, acrylics, and alkyds very effectively. It evaporates at a moderate rate, which allows coatings to form smooth, glossy films without defects like blushing.

Example: In automotive refinishing paints (used by brands like DuPont and PPG Industries), butyl acetate helps control the drying rate and improves gloss and flow of the topcoat.

2. Printing Inks: In the printing industry, butyl acetate is used as a solvent in flexographic and gravure inks for packaging materials, including plastic films and foils. It offers excellent pigment dispersion and consistent viscosity, leading to high-quality print definition.

Example: Ink formulations used in food packaging and plastic labels often rely on butyl acetate for fast yet uniform drying on high-speed printing presses.

3. Adhesives and Sealants: Butyl acetate serves as a solvent in synthetic rubber and polymer-based adhesives, particularly for materials like vinyl and leather. It allows smooth application and rapid bonding due to its balanced evaporation rate.

Example: It is used in shoe manufacturing and furniture lamination adhesives, where strong, flexible bonds are required.

4. Pharmaceuticals and Extraction: In the pharmaceutical sector, butyl acetate is used as a solvent for drug formulation and as an extraction agent for separating active compounds. Its low toxicity and good selectivity make it a safer alternative to chlorinated solvents.

Example: Employed in antibiotic purification and vitamin extraction processes in industrial-scale drug production.

5. Flavors and Fragrances: Because of its fruity, sweet odor (similar to pear or banana), butyl acetate is widely used as a flavoring agent in foods and beverages and as a fragrance component in perfumes and cosmetics.

Example: It is found in fruity perfumes, air fresheners, and flavor essences produced by companies like Givaudan and IFF (International Flavors & Fragrances).

Why Is It an Important Chemical?

Industrial importance:

- It's one of the most widely used esters in the solvent industry.
- Provides an optimal balance of volatility, solvency power, and odor — better than many alcohols or ketones.
- Compatible with a wide range of coating resins (nitrocellulose, acrylics, epoxies, etc.).

Economic significance:

- High-volume production chemical — millions of tons produced annually.
- Essential intermediate in paints, coatings, inks, and fragrance sectors.

Environmental aspect:

- Considered a low-toxicity, biodegradable solvent compared to many alternatives.
- Approved by environmental regulations for use in eco-friendlier coating systems.

PHYSICAL PROPERTIES:

Property	Value / Description
Chemical formula	$C_6H_{12}O_2$
Molar mass	116.16 g/mol
Appearance	Clear, colorless liquid with fruity odor
Odor	Sweet, banana-like or pear-like
Boiling point	~126 °C
Melting point	−78 °C
Density (at 20 °C)	0.882 g/cm ³
Refractive index (n_D²⁰)	1.394
Vapor pressure (at 25 °C)	~10 mm Hg
Flash point	24 °C (closed cup) – highly flammable
Autoignition temperature	421 °C
Viscosity (at 25 °C)	0.73 mPa·s
Solubility in water	Slightly soluble (0.68 g/100 mL at 20 °C)
Miscibility	Miscible with alcohols, ethers, ketones, and hydrocarbons
Vapor density (air = 1)	4.0
Evaporation rate (BuAc = 1)	Defined as 1 (reference solvent in coatings industry)

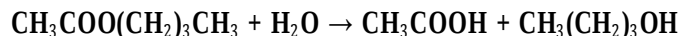
CHEMICAL PROPERTIES:

1. Hydrolysis (Ester cleavage)

In the presence of an acid or base, butyl acetate undergoes hydrolysis to form acetic acid and n-butanol.

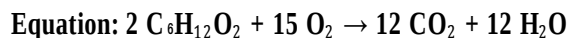
Acidic hydrolysis: reversible reaction.

Basic hydrolysis (saponification): irreversible and faster.



2. Combustion:

Like most organic esters, it burns readily in air to form carbon dioxide and water, releasing energy.

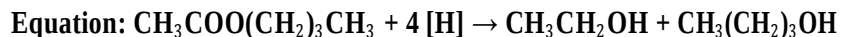


3. Reaction with Strong Oxidizing Agents

Butyl acetate reacts vigorously with oxidizers such as potassium permanganate or nitric acid, leading to possible explosive or exothermic oxidation forming CO_2 , CO, and water. Hence, it should be stored away from oxidizing chemicals.

4. Reduction (laboratory)

Under reducing conditions, esters like butyl acetate can be reduced by LiAlH_4 (lithium aluminium hydride) to form two alcohols: n-butanol and ethanol.



5. Reaction with Ammonia / Amines

When heated with ammonia or amines, butyl acetate can undergo aminolysis to form amides or substituted amides, releasing n-butanol.



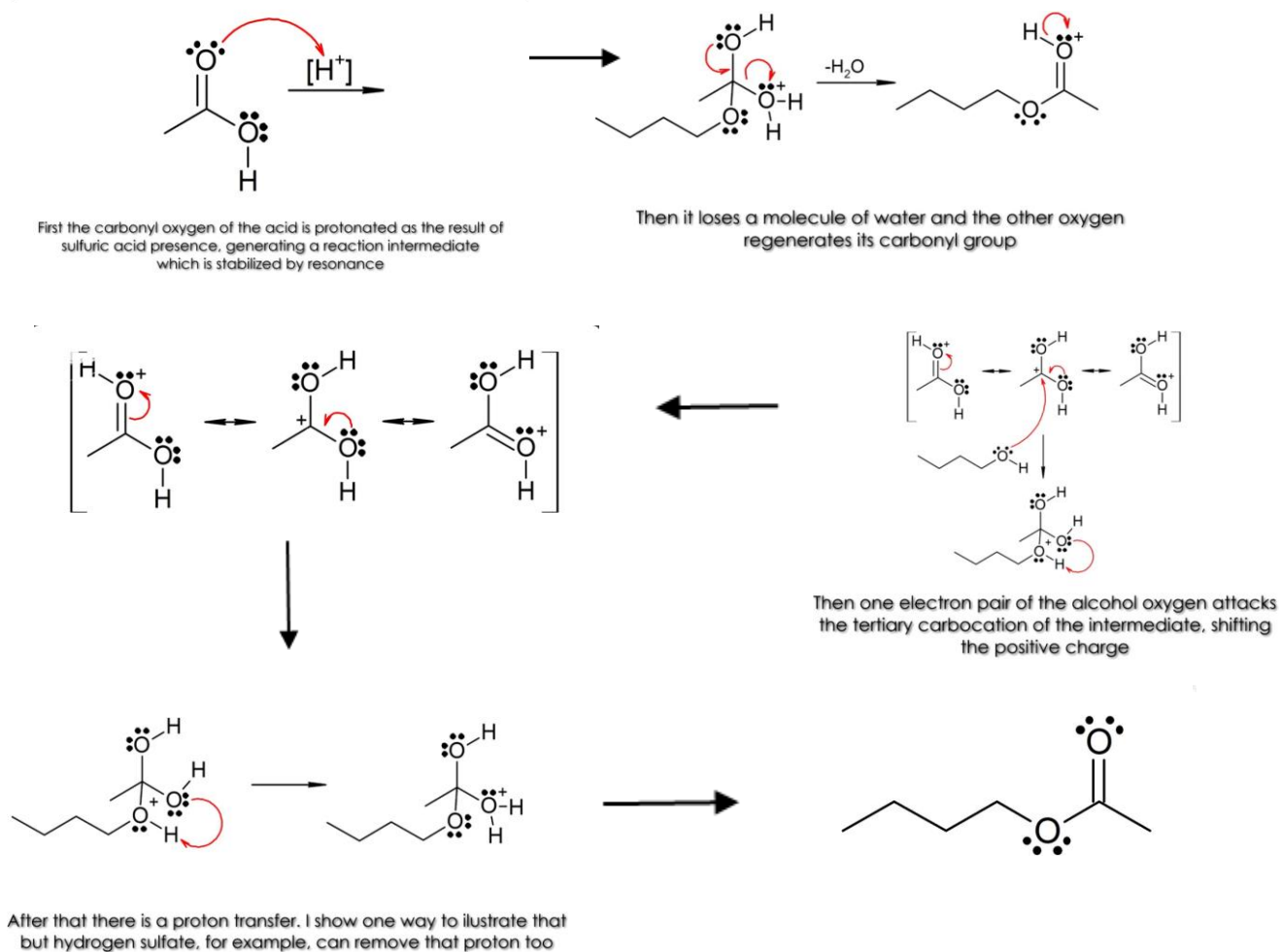
6. Polymer Compatibility

Chemically, butyl acetate is non-reactive toward most resins and polymers, making it ideal as a solvent in paints and coatings. It doesn't chemically alter the resin but aids in film formation.

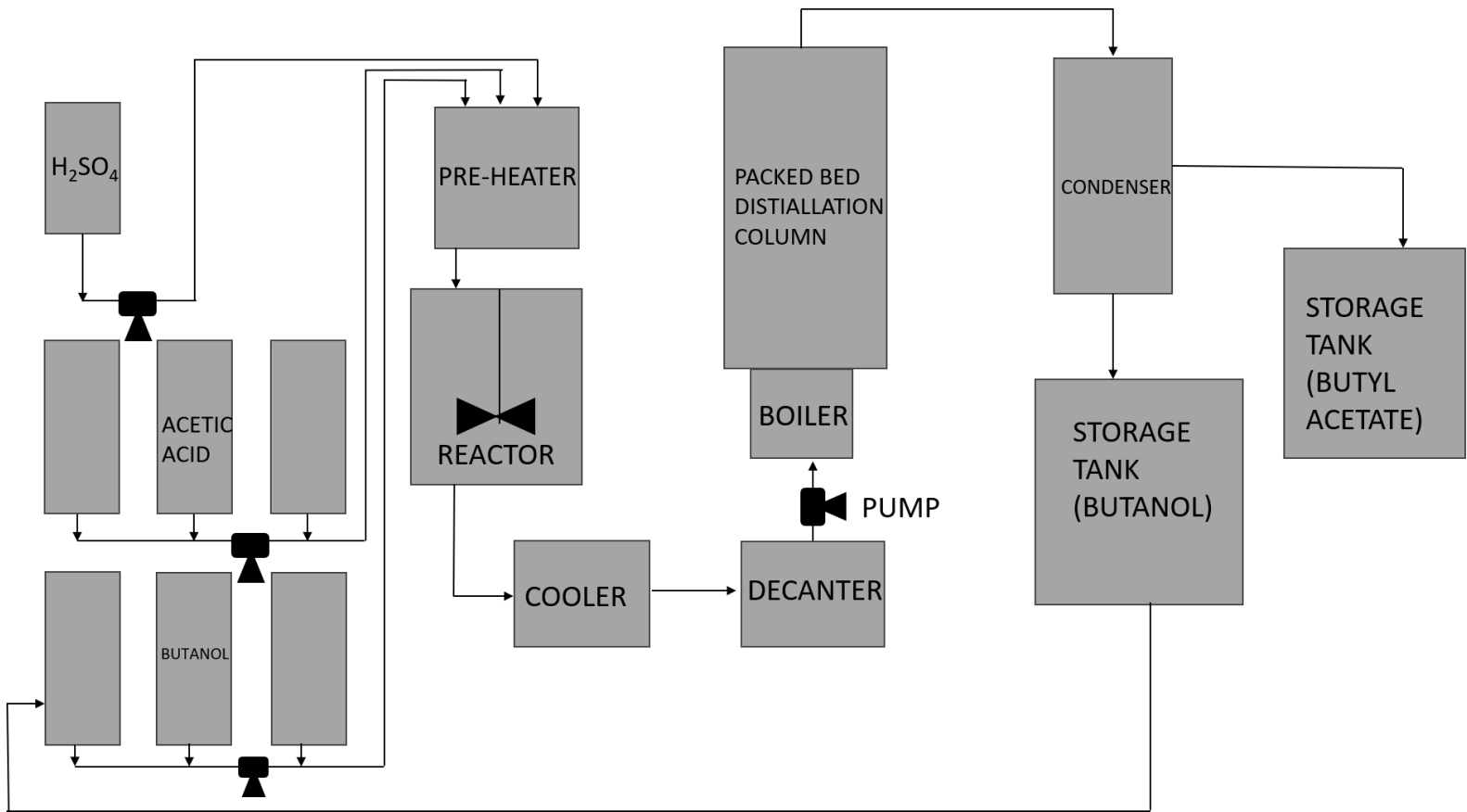
CHEMICAL REACTION & MECHANISM



Mechanism:



PROCESS FLOW DIAGRAM



PROCESS DESCRIPTION

Butan-1-ol, taken 20% in excess of the acetic acid is pumped, along with the latter and sulfuric acid, to a preheater and then to a CSTR. CSTR is Continuous Stirred Tank Reactor. There is an agitator present in CSTR. An esterification reaction occurs at 110-120 degrees Celsius and 1atm pressure. After the reaction, an ester called butyl acetate is produced. Butanol is left over and sulfuric acid too. Water is produced in the reaction and must be removed to avoid hydrolysis of the ester. The mixture leaving the CSTR is cooled to 25 degrees Celsius in a cooler. Then the mixture is sent to a decanter. Here separation occurs on the basis of miscibility. Since butanol and butyl acetate are miscible and less dense than the miscible mixture of water and sulfuric acid, the mixture separates out into two distinct layers. The top layer is drained and the butanol-butyl acetate mixture is then distilled for purification. The mixture is sent to a boiler to boil it and bring the components to boiling point (since distillation is separation on the basis of BP). A packed bed distillation column is used; it is of very large height to ensure accurate separation

MATERIAL BALANCE CALCULATION

Chemical: Butyl acetate

Capacity: 100 tonnes per day= 100000 kg per day, operational 12 hours

Densities used:

- n-Butanol: $\rho_{\text{butanol}} = 810 \text{ kg/m}^3$
- Glacial acetic acid: $\rho_{\text{acid}} = 1049 \text{ kg/m}^3$
- Concentrated H_2SO_4 (98%): $\rho_{\text{H}_2\text{SO}_4} = 1840 \text{ kg/m}^3$
- Butyl acetate (product): $\rho_{\text{prod}} = 882 \text{ kg/m}^3$
- Water: $\rho_{\text{water}} = 1000 \text{ kg/m}^3$

Molar masses used:

- $M_{\text{butyl acetate}} = 116.16 \text{ g/mol}$
- $M_{\text{n-butanol}} = 74.12 \text{ g/mol}$
- $M_{\text{acetic acid}} = 60.05 \text{ g/mol}$
- $M_{\text{H}_2\text{O}} = 18.015 \text{ g/mol}$

Basic Calculations

Butyl acetate produced = 100000 kg

Kilomoles of butyl acetate= 100000/116.16 kmol

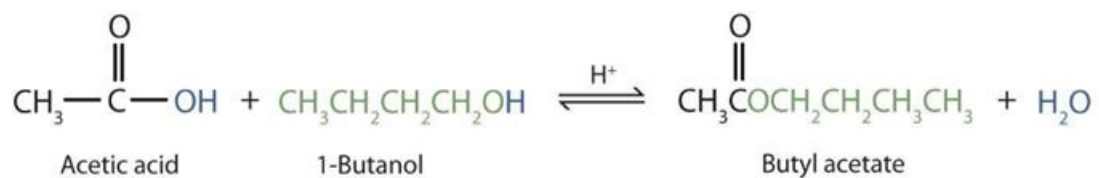
Therefore,

Water produced= 100000/116.16 kmol

These are the kilomoles formed on 67% yield

Kilomoles formed if yield was 100% = 100000/(116.16*0.67) kmol

From the chemical reaction, we see that all stoichiometric coefficients are equal.



Therefore, Acetic acid used = $100000 / (116.16 \times 0.67)$ kmol

Butan-1-ol is taken in excess to shift the equilibrium in the forward direction and promote the formation of butyl acetate.

Assuming **20%** excess,

Butan-1-ol used: $(100000 \times 1.2) / (116.16 \times 0.67)$ kmol

Assuming that the moles of sulfuric acid catalyst used are 0.5% of the moles of butanol,

H₂SO₄ used: $(100000 \times 1.2 \times 0.005) / (116.16 \times 0.67)$ kmol

ENERGY BALANCE CALCULATIONS:

COMPOUNDS	$\Delta H_{F(298K)}$	HEAT CAPACITY
Butanol	-4425.675 KJ/Kg	2309.7 J/Kg-K
Butyl acetate	-4553 KJ/Kg	1937.93 J/Kg-K
Acetic Acid	-8053.66 KJ/kg	2050 J/Kg-K
Water	-15888.88 KJ/Kg	4200 J/Kg-K

CALCULATIONS:

$$\begin{aligned}\text{Acetic Acid : } & -8058.66 + (1 \cdot 2.050 \cdot (390 - 298)) \text{ kJ/kg} \\ & = \mathbf{-7870.06 \text{ kJ/kg}}\end{aligned}$$

$$\begin{aligned}\text{Butanol : } & -4425.675 + (1 \cdot 2.309 \cdot (390 - 298)) \text{ kJ/kg} \\ & = \mathbf{-4213.247 \text{ kJ/kg}}\end{aligned}$$

$$\begin{aligned}\text{Butyl acetate: } & -4533 + (1 \cdot 1.937 \cdot (390 - 298)) \text{ kJ/kg} \\ & = \mathbf{-4374.796 \text{ kJ/kg}}\end{aligned}$$

$$\begin{aligned}\text{Water: } & -15888.88 + (1 \cdot 4.2 \cdot (390 - 298)) \text{ kJ/kg} \\ & = \mathbf{-15502 \text{ kJ/kg}}\end{aligned}$$

$$\text{Heat of reaction: } -15502 - 4374.796 - (-4213.247 - 7870.06) = \mathbf{-7793.969 \text{ kJ/kg}}$$

PLANT DESIGN CALCULATIONS

Assumptions:

- Storage tanks are at ambient temperature and pressure (no pressurised storage).
- Ullage / working headspace: 20% headspace (i.e., gross tank volume = required liquid volume ÷ 0.8).
- Tanks assumed vertical cylindrical; with H/D=1.5.
- Conical roofs:

i) Lowest cost per m³

ii) Ease of fabrication

iii) Good for both volatile and corrosive chemicals

- Relevant lining material with explanation for each tank are also specified.

Storage Tank Design Important Formulae

$$H/D = 1.5$$

$$\text{Thickness of Cylindrical Shell} = P D_i / 2 \sigma J - P$$

$$\text{Thickness of Spherical Roof} = P D_i / 4 \sigma J - P$$

$$\text{Thickness of Conical Roof} = P D_i / 4 \sigma J \sin \theta$$

$$\text{Thickness of Flat Base} = \sqrt{3} \rho_f H A^{2/4} \sigma J$$

where,

- P is the absolute pressure in the tank
- D_i is the internal diameter of the tank
- H is the height of the tank
- A is the cross-sectional area of the tank
- σ is the yield stress of the construction material
- J is the joint efficiency (generally 85%) θ is the semi-vertical angle of cone (45° recommended)

Tank for butyl acetate: Carbon steel

Nitrogen blanketing due to highly flammable nature. PTFE lining.

Assuming butyl acetate is stored for 7 days,

Mass of butyl acetate = $7 \times 100000 \text{ kg}$

Volume of butyl acetate = $700000/882 \text{ m}^3$

Tank volume accounting for headspace = $(700000/882)/0.8$

Volume of cylindrical tank = $\pi D^2 H/4 = \pi D^3 \times 1.5/4$

Therefore tank diameter, $D = 9.443 \text{ metres}$

And height $H = 14.165 \text{ metres}$

$P = \rho \times g \times H + \text{atmospheric pressure} = 122561.5293 + 101325 = 223886.5293 \text{ Pa}$

Assuming σ for carbon steel is 140 MPa ,

Thickness of cylindrical shell = $P D_i/2 \sigma J - P = 8.891 \text{ mm}$

Roof thickness = $P D_i/4 \sigma J \sin \theta = 6.281 \text{ mm}$

Base thickness = $\sqrt{3} \rho_f H A^2/4 \sigma J = 621.455 \text{ mm}$

Tanks for butan-1-ol: Carbon steel

Non-corrosive but flammable, so flame arrestor and nitrogen blanketing needed. PTFE lining.

Assuming butan-1-ol is stored for 7 days,

Mass of butan-1-ol = $114283.9521 \times 7 \text{ kg}$

Volume of butan-1-ol = $114283.9521 \times 7/810 \text{ m}^3$

Tank volume accounting for headspace = $(114283.9521 \times 7/810)/0.8$

Dividing this volume into 3 identical tanks,

$$\text{Volume per tank} = \pi D^2 H / 4 = \pi D^3 * 1.5 / 4$$

Therefore tank diameter, $D=7.043$ metres

And height $H=10.564$ metres

$$P = \rho * g * H + \text{atmospheric pressure} = 185267.6004 \text{ Pa}$$

Assuming σ for carbon steel is 140MPa,

$$\text{Thickness of cylindrical shell} = P D_i / 2 \sigma J - P = 5.487 \text{ mm}$$

$$\text{Roof thickness} = P D_i / 4 \sigma J \sin \theta = 3.877 \text{ mm}$$

$$\text{Base thickness} = \sqrt{3} \rho f H A^2 / 4 \sigma J = 286.1 \text{ mm}$$

Tanks for acetic acid: Carbon steel

Moderately corrosive, corrosion resistant lining required. PTFE or glass lining.

Assuming acetic acid is stored for 7 days,

$$\text{Mass of acetic acid} = 77158.11439 * 7 \text{ kg}$$

$$\text{Volume of acetic acid} = 77158.11439 * 7 / 1049 \text{ m}^3$$

$$\text{Tank volume accounting for headspace} = (77158.11439 * 7 / 1049) / 0.8$$

Dividing this volume into 3 identical tanks,

$$\text{Volume per tank} = \pi D^2 H / 4 = \pi D^3 * 1.5 / 4$$

Therefore tank diameter, $D=5.668$ metres

And height $H=8.502$ metres

$$P = \rho * g * H + \text{atmospheric pressure} = 188816.4464 \text{ Pa}$$

Assuming σ for carbon steel is 140MPa,

$$\text{Thickness of cylindrical shell} = P D_i / 2 \sigma J - P = 4.500 \text{ mm}$$

$$\text{Roof thickness} = P D_i / 4 \sigma J \sin \theta = 3.180 \text{ mm}$$

$$\text{Base thickness} = \sqrt{3} \rho_f H A^2 / 4 \sigma J = 189.171 \text{ mm}$$

Tanks for sulfuric acid: Carbon steel

Internal environment must remain dry; tank lined with PTFE. Glass lining or PTFE.

Assuming sulfuric acid is stored for 7 days,

$$\text{Mass of sulfuric acid} = 755.5199211 * 7 \text{ kg}$$

$$\text{Volume of sulfuric acid} = 755.5199211 * 7 / 1840 \text{ m}^3$$

$$\text{Tank volume accounting for headspace} = (755.5199211 * 7 / 1840) / 0.8$$

$$\text{Volume of per tank} = \pi D^2 H / 4 = \pi D^3 * 1.5 / 4$$

$$\text{Therefore tank diameter, } D = 1.450 \text{ metres}$$

$$\text{And height } H = 2.175 \text{ metres}$$

$$P = \rho * g * H + \text{atmospheric pressure} = 140584.62 \text{ Pa}$$

Assuming σ for carbon steel is 140 MPa,

$$\text{Thickness of cylindrical shell} = P D_i / 2 \sigma J - P = 0.857 \text{ mm}$$

$$\text{Roof thickness} = P D_i / 4 \sigma J \sin \theta = 0.606 \text{ mm}$$

$$\text{Base thickness} = \sqrt{3} \rho_f H A^2 / 4 \sigma J = 8.293 \text{ mm}$$

TANK CALCULATION TABLE

Sr No.	Chemical Stored	Capacity	Dimensions	Thickness
1	Sulfuric Acid	2.872460 m ³	D=1.450m H=2.175m	Shell =0.857mm Roof =0.606mm Base =8.293mm
2	Acetic Acid	646.6 m ³	D=5.668m H=8.502m	Shell =4.500mm Roof =3.180mm Base =189.171mm
3	Butanol	1234.54875 m ³	D=7.043m H=10.564m	Shell =5.487mm Roof =3.877mm Base =286.1mm
4	Butyl Acetate	992.0625 m ³	D=9.443m H=14.165m	Shell =8.891mm Roof =6.281mm Base =621.455mm

REACTOR DESIGN

(ALL CALCULATIONS AS PER BATCH)

MATERIAL OF CONSTRUCTION : SS316L stainless steel

Reason: It is resistant to acetic acid, butanol, ester product, and dilute sulfuric acid catalyst.

Yield stress = **170 MPa**

REACTANTS	QUANTITY PER BATCH	DENSITY @STP	VOLUME (m ³)
Acetic Acid	2571.94 Kg	1055 Kg/m ³	30.6474 m ³
Butanol	3809.46 Kg	810 Kg/m ³	58.788 m ³
Sulfuric Acid	25.18 Kg	1800 Kg/m ³	0.13686 m ³

CONDITIONS :

TEMPERATURE : 110°C

PRESSURE : 1 atm

DENSITIES (@ 110°C and 1 atm pressure):

Acetic Acid = 920 Kg/m³

Butanol = 720 Kg/m³

Sulfuric Acid = 1620 Kg/m³

MIXTURE DENSITY = 880 Kg/m³

TOTAL VOLUME : 30.6474 m³ + 58.788 m³ + 0.13686 m³ = 89.57226 m³

$$H/D = 1$$

Therefore, **D = 4.8494 m**

$$H = 4.8494 \text{ m}$$

Considering safety factor (25%), **H = 2.083*1.25 = 6.0618 m**

Area = 18.4699 m²

Pressure = 101.325 KPa + (H * rho * g)

$$= 101.325 + (6.0618 * 880 * 9.81 * 10^{(-3)}) \text{ KPa}$$

$$= 153.6553 \text{ KPa}$$

Shell Thickness : = **$P D_i / 2 \sigma J - P$**

$$t = 153655.3 \times 4.8494 / 2 \times 170000000 \times 0.85 - 153655.3$$

$$t = 0.00257 \text{ m} = 2.57 \text{ mm}$$

Roof Thickness : $= P D_i / 4 \sigma J - P$

$$t(r) = 153655.3 \times 4.8494 / 4 \times 170000000 \times 0.85 - 153655.3$$

$$t(r) = 0.001289 \text{ m} = \mathbf{1.289 \text{ mm}}$$

Base Thickness : $= \sqrt{(3 \rho_f H A^2 / 4 \sigma J)}$

$$t(b) = \sqrt{(3 \times 880 \times 6.0618 \times 341.139 / 4 \times 170000000 \times 0.85)}$$

$$t(b) = 0.09718 \text{ m} = \mathbf{97.186 \text{ mm}}$$

AGITATOR

$$D_A / D_T = 1/3$$

$$D_T = 4.8494 \text{ m (as calculated above)}$$

$$\text{Hence, } D_A = 1/3 \times 4.8494 = \mathbf{1.616 \text{ m}}$$

$$E / D_T = 1/3$$

$$\mathbf{E = 1.616 \text{ m}}$$

$$W / D_A = 1/5$$

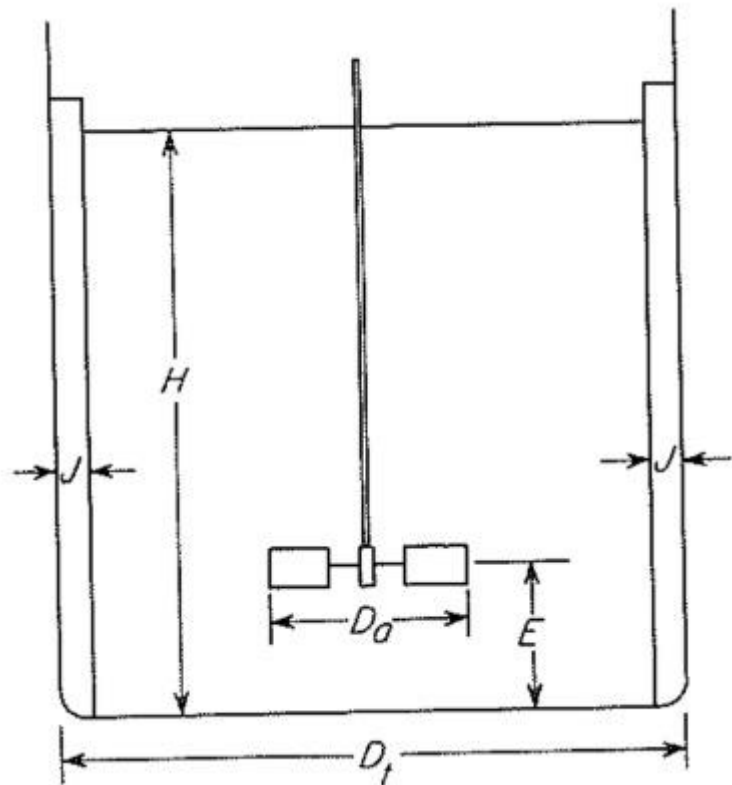
$$\mathbf{W = 0.323 \text{ m}}$$

$$L / D_A = 1/4$$

$$\mathbf{L = 0.404 \text{ m}}$$

$$J / D_T = 1/12$$

$$\mathbf{J = 0.404 \text{ m}}$$



POWER CONSUMED

$$P = K_T \cdot N^3 \cdot D^5 \cdot \rho$$

Assuming, 45° pitch blade turbine impeller having three blades propeller is used for axial flow and better mixing and turns 100rpm= 1.667 rps

$$K_T = 0.87$$

$$P = 0.87 \times 1.667^3 \times 1.616^5 \times 880 = 39085.4851 \text{ Watt} = \mathbf{39.085 \text{ kW}}$$

Time of operation per batch= 4 hr

$$\mathbf{ENERGY \text{ CONSUMPTION} = 39.085 \times 4 = 156.34 \text{ KWh}}$$

PIPING AND PUMPING

ASSUMPTIONS : Diameter of pipe from butanol storage tank to reactor = **4 inch**

(We are using DN40 pipe)

Velocity of flow = **2 m/s**

1) BUTANOL STORAGE TANK TO REACTOR

Volume of butanol entering reactor = 58.788 m³

Diameter of pipe = 4 inch = 0.1016 m

$$A = Q/V$$

Therefore, $Q = A * V$

$$Q = 0.016 \text{ m}^3/\text{sec}$$

$$\text{Time} = 58.788 \text{ m}^3 / 0.016 \text{ m}^3/\text{sec} = 3674.25 \text{ seconds} = 61.2375 \text{ minutes}$$

$$\text{Reynold's number} = \rho * v * d / \text{viscosity} = 137160$$

So, the flow through pipe is fully turbulent !!

Considering the furthestmost tank ,

Length of pipe: 26.2 m

Diameter of pipe : 0.1016 m

Friction factor from Moody diagram= 0.0205

$$\text{Major loss : } f * l/d * u^2/2g$$

$$= 1.077 \text{ m}$$

$$\text{Minor loss: } K * u^2/2g \text{ (considering 3 bends of } 90^\circ \text{ and 3 gate valve wide open)}$$

$$= 0.672 \text{ m}$$

$$H_f = 1.75 \text{ m}$$

Applying Bernoulli equation,

$$\Delta z = H_p - H_f$$

$$H_p = 8.0118 \text{ m (considering } \Delta z = \text{height of reactor (almost))}$$

$$\text{Power} = H_p * \text{mass flow rate} * g = 1018.6 \text{ W}$$

$$\text{Considering 85\% efficiency, power consumed} = 1198.354 \text{ W}$$

But, Preheater installed also consumes power roughly around 0.07744 kW

Therefore , **Actual Power Consumed = 1.275 kW**

NET POWER SUCTION HEAD :

$$\text{NPSHa} = H_{\text{atm}} + h(\text{static}) - H_{\text{vap}} - h(\text{loss,static})$$

So, **NPSHa (calculated) = 23.6473**

So, PUMP DETAILS :

PUMP MOTOR RATING = 1.5 kW (=2HP)

Required NPSHr: ≤ 3 m

2) ACETIC ACID STORAGE TANK TO REACTOR

Volume of Acetic Acid entering reactor = 30.6474 m³

Time = 3674.25 seconds

$$Q = 30.6474 \text{ m}^3 / 3674.25 \text{ s} = \mathbf{0.0083411 \text{ m}^3/\text{s}}$$

$$V = 2 \text{ m/s}$$

Using, $A = Q/V$

$$D = 0.07287 \text{ m} = 2.8689 \text{ inches}$$

Therefore , we take 2.5 inch standard pipe

We keep the flow rate same , then the velocity of flow changes.

$$\mathbf{V = 2.6338 \text{ m/sec}}$$

$$\mathbf{\text{Reynold's number} = \rho * v * d / \text{viscosity} = 151897.2889}$$

So, the flow through pipe is fully turbulent !!

PUMP CALCULATIONS :

Length of pipe: 21.5 m

Diameter of pipe : 0.0635 m

Friction factor from Moody diagram= 0.0204

Major loss : $f * l/d * u^2/2g$

$$= 2.442 \text{ m}$$

Minor loss: $K * u^2/2g$ (considering 2 bends of 90° and 1 gate valve wide open)

$$= 0.1166 \text{ m}$$

$$H_f = 2.5586 \text{ m}$$

Applying Bernoulli equation,

$$\Delta z = H_p - H_f$$

$H_p = 8.8186 \text{ m}$ (considering Δz = height of reactor (almost))

$$\text{Power} = H_p * \text{mass flow rate} * g = 756.9569 \text{ W}$$

Considering 85% efficiency, power consumed= 890.5376 W

But, Preheater installed also consumes power roughly around 0.0802 kW

Therefore , **Actual Power Consumed = 0.97 kW**

NET POWER SUCTION HEAD :

$NPSH_a = H_{atm} + h(\text{static}) - H_{vap} - h(\text{loss,static})$

So, **NPSHa (calculated) = 17.1082**

So, PUMP DETAILS :

PUMP MOTOR RATING =1.5 kW (2 HP)

Required NPSHr: ≤ 3 m

2) SULFURIC ACID STORAGE TANK TO REACTOR

Volume of H₂SO₄ entering reactor = 0.13686 m³

Time = 3674.25 seconds

Q = 0.13686 m³/3674.25 s = 0.00003724 m³/s

V = 2 m/s

Using, $A = Q/V$

D = 0.004869 m = 0.1916 inches

Therefore , we take 0.25 inch standard pipe

We keep the flow rate same , then the velocity of flow changes.

V = 1.1759 m/sec

Reynold's number = $\rho \cdot v \cdot d / \text{viscosity} = 577.277$

So, the flow through pipe is laminar !!

PUMP CALCULATIONS :

Length of pipe: 24.5 m

Diameter of pipe : 0.00635 m

Friction factor from moody diagram= 0.110

Major loss : $f \cdot l/d \cdot u^2/2g$

= 30.146 m

Minor loss: $K \cdot u^2/2g$ (considering 2 bends of 90° and 1 gate valve wide open)

= 0.2325 m

$H_f = 30.3785$ m

Applying Bernoulli equation,

$$\Delta z = H_p - H_f$$

$$H_p = 36.638 \text{ m (considering } \Delta z = \text{height of reactor (almost))}$$

$$\text{Power} = H_p * \text{mass flow rate} * g = 24.6283 \text{ W}$$

$$\text{Considering 85\% efficiency, power consumed} = 28.9745 \text{ W}$$

But, Preheater installed also consumes power roughly around 0.04 kW

Therefore, **Actual Power Consumed = 0.06897 kW**

NET POWER SUCTION HEAD :

$$\text{NPSHa} = H_{\text{atm}} + h(\text{static}) - H_{\text{vap}} - h(\text{loss, static})$$

$$\text{So, NPSHa (calculated)} = 12.135$$

So, PUMP DETAILS :

$$\text{PUMP MOTOR RATING} = 0.12 \text{ kW (0.16 HP)}$$

Required NPSHr: $\leq 3 \text{ m}$

4) REACTOR TO DISTILLATION COULMN

Assumptions: Diameter of pipe is 4 in = 0.1016 m

Here,

Butanol, Butyl acetate, H₂SO₄ and water are going through a single pipe.

Volume of butanol: 15.52 m³

Volume of H₂O: 5.170 m³

Volume of Butyl acetate: 37.793 m³

Volume of H₂SO₄ : 0.137 m³

Volumetric flow rate : 0.016 m³/s

Viscosity of mixture = **10⁻³ Pa-s**

Density of mixture = $(1/\rho = \sum w_i/\rho_i) = \mathbf{880.5 \text{ Kg/m}^3}$

Reynolds Number: 1789176 (TURBULENT FLOW)

Length of pipe: 22.05 m

Diameter: 0.1016 m

Velocity = 1.973 m/s

Friction factor from Moody diagram = 0.0167

Major loss = 0.719 m

Minor loss = 0.2579 m

$H_f = 0.9769$ m

Applying Bernoulli equation,

$\Delta z = H_p - H_f$

$H_p = 4.9769$ m

Power of pump = $h_p \times \text{mass flow rate} \times g$

Power = 687.823 W

Considering 85 % efficiency $\implies$ power consumed = 809.204 W

A cooling tower is installed just before distillation column , so the pressure drop due to cooling tower = 0.15 kw

Therefore , **Actual Power Consumed = 0.959 kW**

So, PUMP DETAILS :

PUMP MOTOR RATING =1.5 kW (2 HP)

Required NPSHr: ≤ 3 m

DECANTER

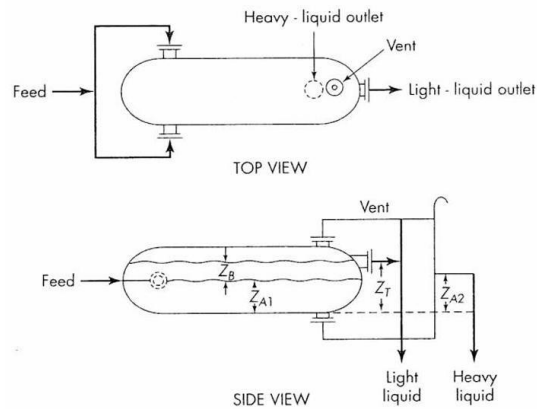


FIGURE 2.5
Continuous gravity decanter for immiscible liquids.

$$Z_B \rho_B + Z_{A1} \rho_A = Z_{A2} \rho_A$$

$$Z_{A1} = Z_{A2} - Z_B \frac{\rho_B}{\rho_A} = Z_{A2} - (Z_T - Z_{A1}) \frac{\rho_B}{\rho_A}$$

The top layer will be organic. Butanol+ butyl acetate

The bottom layer will be water+ H₂SO₄

Volume of substances leaving CSTR:

- Butyl acetate = 37.793 cubic metres
- Butan-1-ol = 15.52 cubic metres
- Water = 5.170 cubic metres
- H₂SO₄ = 0.137 cubic metres

Since densities have been previously defined, we can calculate the effective density of top (Z_B) and bottom layers (Z_{A1})

The initial component masses are:

Butyl acetate= volume*density= 37.793 m³ * 882 kg/m³ = 33333.43 kg

Similarly for other components,

Butanol= 12571.20 kg

Water= 5169.60 kg

H₂SO₄= 258.08 kg

$$\rho_{\text{effective}} = \frac{\sum M_i}{\text{Total Layer Volume}}$$

For top layer, $\rho_B = 45904.64/53.313 = 861.04 \text{ kg/m}^3$

For bottom layer, $\rho_A = 5421.68/5.307 = 1021.61 \text{ kg/m}^3$

The **continuous phase** in a liquid-liquid separation system, such as a decanter, is typically the phase that has the **highest volume** and/or the phase that **wets the walls** of the vessel.

Hence, the continuous phase will be the **Top (Organic) Layer**.

Dimensions and design:

Horizontal cylinder with elliptical heads

The standard length to diameter ratio is usually 2 to 5. Let's take 2. Also taking a volume allowance of 10%.

Total volume = volume of horizontal cylinder + volume of 2 elliptical heads of (2:1 dimension)

$$= \pi D^2 L / 4 + \pi D^3 / 12 = \pi D^3 ((2/4) + (1/12))$$

$$\text{Therefore, } 58.62 / 0.9 = \pi D^3 ((2/4) + (1/12))$$

$$D = 3.288 \text{ m}$$

$$\text{Thus, } L = 6.576 \text{ m}$$

Thickness of wall and heads:

Material: Stainless Steel (SS)

- Use: The most standard and frequently used material in chemical decanters, especially grade SS 316.

Parameter	Value (Metric)	Source/Reasoning
Design Pressure (P)	0.5 MPa	Typical design value for decanter safety margin
Allowable Stress (S)	138 MPa	ASME Code value for SS 316 at 100 degrees Celsius.
Weld Efficiency (E)	0.85	Assumed standard spot-examined butt weld
Corrosion Allowance (C)	2mm or 0.002m	Standard allowance for mild/moderate corrosion

A. Thickness of Cylindrical Shell Wall

This thickness is calculated based on the **hoop stress** equation (circumferential stress) for thin-walled cylinders:

$$t_{\text{shell}} = \frac{PR}{SE - 0.6P} + C$$
$$= 9.026\text{mm}$$

B. Thickness of 2:1 Elliptical Heads

The thickness for the elliptical head is calculated based on the formula derived from the bending stress analysis:

$$t_{\text{head}} = \frac{PD}{2SE - 0.2P} + C$$
$$= 9.011\text{mm}$$

Heights of layers

Total tank volume= 58.62/0.9 m³

Volume of Two 2:1 Elliptical heads= 9.33 m³

Volume of horizontal cylinder= 55.8 m³

Volume of bottom layer (water+ H₂SO₄)= 5.301 m³

If bottom layer occupies 5.301 m³ of full decanter 58.62/0.9 m³

Then it will occupy x m³ of cylindrical volume 55.8 m³

By cross multiplication, volume of decanter occupied by bottom layer is 4.541 m³.



To determine the height h, we use

$$V_{\text{segment}} = L \cdot \left[R^2 \cos^{-1} \left(\frac{R-h}{R} \right) - (R-h) \sqrt{2Rh - h^2} \right]$$

Equating V_{segment} to 4.541 m^3 , we get height h as $0.435\text{m} = Z_{A1}$

Similarly when we equate V_{segment} to 58.62 m^3 in the above formula, we obtain total height of the liquid in decanter = 3.178m .

$$Z_B = 3.178 - 0.435 = 2.743\text{m}$$

Then finding height of the outlet to remove top layer of butanol+ butyl acetate, Z_{A2}

$$Z_B \rho_B + Z_{A1} \rho_A = Z_{A2} \rho_A$$

$Z_{A2} = 2.747\text{m}$ measured from the bottom of the decanter.

Time required for separation

Since the continuous phase is the organic layer i.e. top layer, effective viscosity is 1.04 cP

$$t = \frac{100\mu}{\rho_A - \rho_B}$$

$$= 100 * 1.04 / (1021.61 - 861.04) = 0.648\text{h} = 38.88\text{min}$$

PACKED BED DISTILLATION COLUMN

Liquid entering:

Butyl acetate= 37.793 m^3 ; 124 degrees Celsius

Butan-1-ol= 15.52 m^3 ; 118 degrees Celsius

Total volume= 53.313 m^3

Since the mixture is non-corrosive (esters and alcohols), the most common and cost-effective choice is:

Material: Carbon Steel (CS)

Let's assume a typical commercial random packing, like **1-inch Ceramic Raschig Rings** for the following calculations, as they are a common reference.

Packing Parameter	Value (Example: 1" Raschig Ring)	Symbol
Nominal Size	25.4mm	d_p
Sphericity (for calc)	Approximately 0.4	Φ_s
Void Fraction	Approximately 0.73	ϵ
Specific Surface Area	$185 \text{ m}^2/\text{m}^3$	a_p

We take the diameter to be 1.5m and length of distillation column to be 35m. This gives us an L/D ratio of 23.333 which is within the industrial bounds (max height being 175ft or 53m according to the Rule Book for Chemical Engineers).

From the diameter we can compute area= 1.767 m^2

Packing Length (Height)

The total packing height (H_T) is the number of **Theoretical Stages (N_T)** multiplied by the **Height Equivalent to a Theoretical Plate (HETP)**.

A. Number of Stages (N_T)

N_T is determined by the **VLE curve**, feed composition, and desired product purity using the **McCabe-Thiele Method** or **Fenske Equation**.

For high purity, no. of stages required are higher. Plus, the boiling points are close so more theoretical stages are required for separation.

Let's take $N_T=45$

B. HETP Estimation

The HETP for random packing is typically estimated based on the column diameter and packing size.

$$\text{HETP} = 0.7\text{m}$$

C. Total Packed Length (H_T)

This is a very tall column, which is ideal for such a difficult separation.

Total length of packing is $0.7 \times 45 = 31.5\text{m}$

We must either assume operation time or superficial velocity to proceed further.

Assuming operational time to be 2 hours,

Empty tower velocity/ superficial velocity is

$$35/2 = 17.5 \text{ m/h}$$

$$\text{Channel velocity} = 17.5 / \epsilon = 23.973 \text{ m/h}$$

$$\frac{\Delta p}{L} = \frac{150 \bar{V}_0 \mu (1 - \epsilon)^2}{\Phi_s^2 D_p^2 \epsilon^3} + \frac{1.75 \rho \bar{V}_0^2 (1 - \epsilon)}{\Phi_s D_p \epsilon^3}$$

This is Ergun equation

Delta P is the pressure drop of the packed bed

L is the height of packed bed = 31.5m

V_0 is the superficial or empty column velocity = 17.5 m/h

μ is the effective viscosity of butan-1-ol and butyl acetate = 0.000229 Pa.s

D_p is the particulate diameter of packing material = 25.4mm

Φ_s is 0.4

ϵ is 0.73

ρ is the effective density of butan-1-ol and butyl acetate = 856.96 kg/ m³

Therefore, pressure drop per unit length comes out to be 0.9004 Pa/m

Since packed bed height is 31.5m, total pressure drop is 28.36 Pa.

Thus a pump supplying this pressure must be attached before the liquid enters distillation column.

SAFETY PRECAUTIONS

Ensuring safety during operations such as packed bed distillation and the handling of chemicals like **butyl acetate** is essential. The following safety measures combine general process-safety guidelines with material-specific precautions.

1. Personal Protective Equipment (PPE)

All personnel must wear proper PPE, including:

- Safety goggles or face shields
- Chemical-resistant gloves
- Flame-resistant lab coats or overalls
- Closed footwear
- Respiratory protection where vapors may exceed permissible limits

PPE helps minimize exposure to hazardous chemicals and prevents skin/eye injuries.

2. Training and Supervision

Operators should receive thorough training on:

- Standard operating procedures
- Equipment handling
- Potential hazards related to packed bed distillation
- Correct response during emergencies

Only trained personnel should carry out critical operations.

3. Adequate Ventilation

Butyl acetate vapors are irritating and flammable; therefore:

- Ensure proper general ventilation
- Use **local exhaust systems** near sources of vapor generation
- Prevent vapor accumulation in confined spaces

Good ventilation minimizes inhalation risks and reduces fire hazards.

4. Instrumentation and Monitoring

Install and maintain instruments for continuous monitoring of:

- Temperature
- Pressure
- Flow rates
- Vapor concentrations (if required)

Early detection of abnormal conditions helps prevent accidents.

5. Chemical Compatibility

Check that all materials in contact with process fluids (packing, gaskets, piping) are:

- Chemically compatible with butyl acetate and other chemicals used
- Resistant to corrosion and solvent attack

Using incompatible materials can lead to leakage, corrosion, or system failure.

6. Emergency Shutdown Procedures

Prepare clear and accessible emergency shutdown steps.

All personnel should:

- Know how to stop pumps, heaters, or reactors safely
- Understand evacuation routes
- Be trained in using fire extinguishers, emergency showers, and eyewash stations

Quick action reduces the severity of incidents.

7. Pressure Relief and Fire Safety

- Use appropriate pressure-relief devices to prevent over-pressurization.
- Test and inspect these devices regularly.
- Keep fire extinguishers (Class B for flammable liquids) nearby.
- Maintain emergency showers and eyewash stations in working condition.

Butyl acetate is a **Class IB highly flammable liquid**, so strict fire-prevention practices are required.

8. Lockout/Tagout and Equipment Isolation

Before maintenance:

- Shut down equipment completely
- Isolate energy sources
- Apply lockout/tagout procedures

This prevents accidental start-up and ensures worker safety.

9. Hazardous Area Classification

Identify areas where flammable vapors may be present.

Implement controls such as:

- Explosion-proof electrical fittings
- Spark-proof tools
- Proper grounding and bonding to avoid static discharge

This is especially important because butyl acetate vapors can ignite easily.

10. Material-Specific Safety (Butyl Acetate)

Flammability

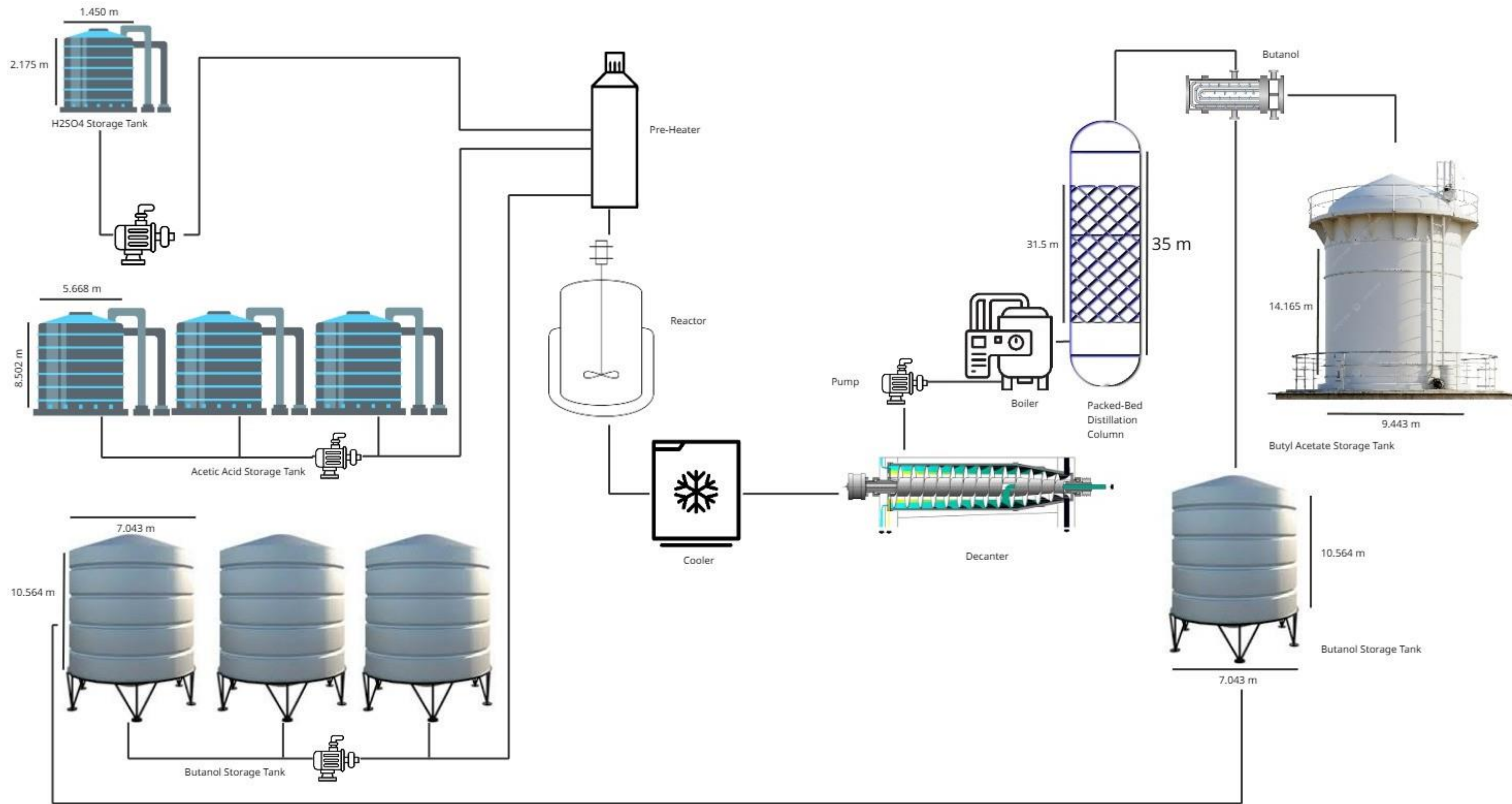
- Highly flammable; vapors can ignite at low temperatures.
- Eliminate ignition sources and ensure proper grounding to avoid static sparks.

Health Hazards

- Vapors may cause respiratory irritation, dizziness, nausea, and headache.
- Skin exposure can cause irritation or dermatitis.

Precaution:

Ensure proper ventilation and minimize exposure through PPE (gloves, goggles, masks).



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